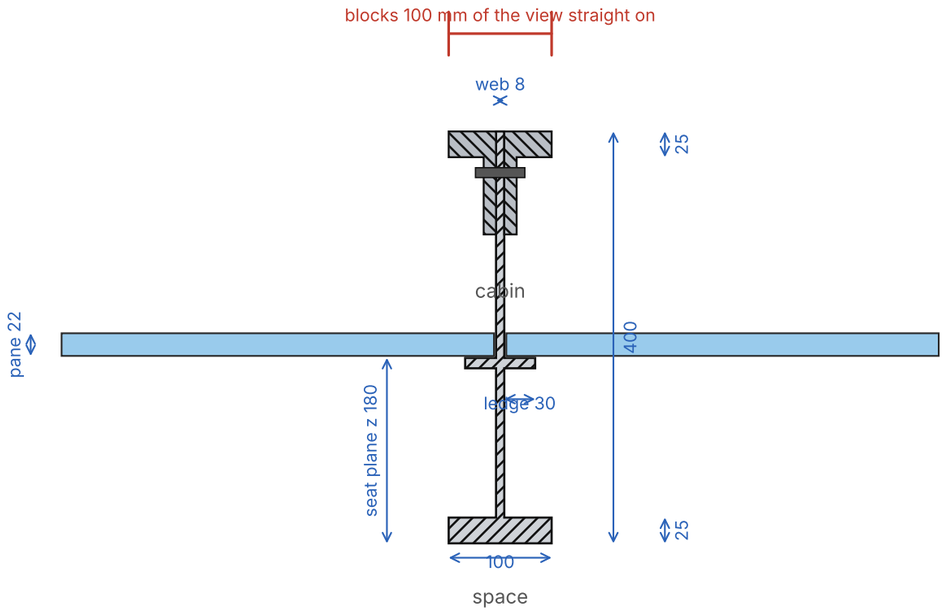
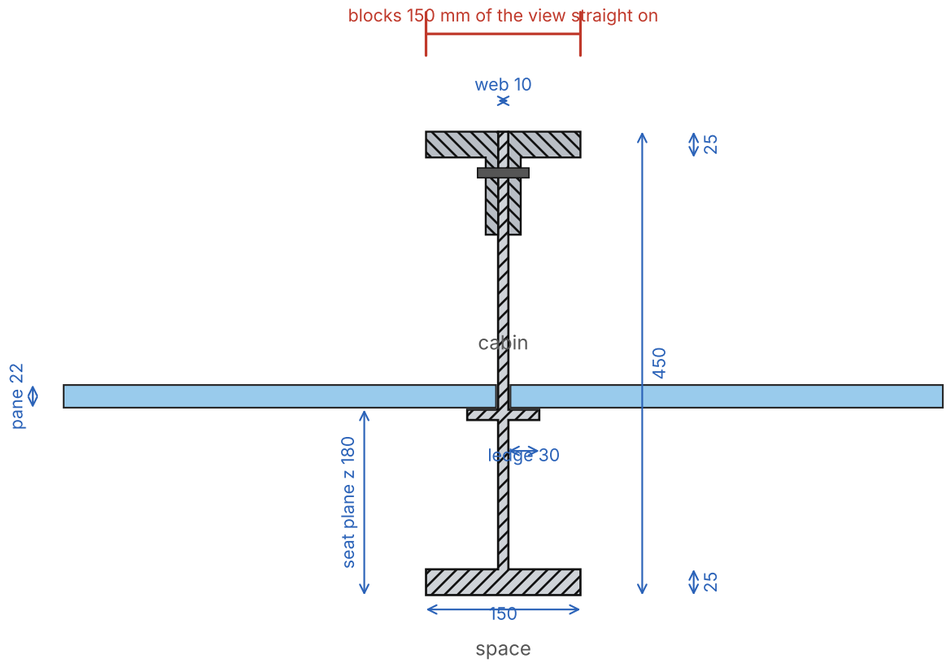
HCSA sheet RE-AB (2 Nov 2026, PROPOSAL): main rib option A against option B, section through the main rib at mid-span with the panes seated

Revised 2 Nov afternoon with the real station loads along the seam (see 'The seam with the real station loads'): the seals open first at the hardpoints, not in the bays. Both options keep the I 180 × 60 × 10 secondary ribs, the 22 mm BOROFLOAT panes at 531 mm on the z 180 seat plane, the 1
Only the six main ribs per hexagon (and the five per pentagon), their cap angles and the hub core change. Numbers from the whole-station shell model (run_window_check.py) and the 2 Nov rib study;
the corner, pane and seam numbers are at 1 atm unless said otherwise. Limit stress 145 MPa (factor 2.0 on 6061-T6 ultimate), yield 240 MPa, fitting factor 1.15 on the seams.

Option B, thin: main rib I 400 × 100 × 25, web 8 (P-04, P-04A)
yields first at 1.96 atm as the hexagon main rib; the seals first reach 0.11 mm at the hardpoints at 2.9 atm, a full atmosphere later



Option A, leak before break kept: main rib I 450 × 150 × 25, web 10 (P-10, P-10A)
the frame yields first at 2.92 atm; the seals first reach 0.11 mm at the hardpoints at 3.4 atm, close behind, so a warning needs a d



	Today (Rev E, 24 Oct)	Option B, thin	Option A, leak before break
Main rib section	I 400 × 150 × 20, web 8	I 400 × 100 × 25, web 8	I 450 × 150 × 25, web 10
Secondary rib	AA I 6 × 4.03 (152 × 102)	I 180 × 60 × 10, web 5	I 180 × 60 × 10, web 5
Open glass, straight on (exact drawing)	43 %	61 %	57 %
Open glass at 23 degrees (shell model)	22 %	33 %	29 %
Worst member at 1 atm, MPa (limit 145)	hex main rib 109	hex main rib 123	frame 82 (main rib 82)
First yield, atm	2.20 (main rib)	1.96 (main rib)	2.92 (frame)
Seals first reach 0.11 mm (at the hardpoints, real s	about 3.0 atm	2.9 atm	3.4 atm
Seals open before the metal yields?	no	no (rib 1.96, seals 2.9)	no, but close (frame 2.92, seals 3.4); a designed fuse would fix it
Pane stress s1 at 1 atm, MPa / glass odds	9.3 / 91 %	9.5 / 91 %	8.5 / 93 %
Hexagon bow at 1 atm, mm	0.39	0.52	0.40
Corner block load at 1 atm, kN	283	281	306
Structure mass, t (shell model)	38.6	36.7	40.5
Main rib + two cap angles, kg per m	about 27	29.2	39.1
Launch cost of the difference vs B		0	+3.8 t: about 0.8 to 11 M dollars at 200 to 3,000 dollars per kg
Hub core height / fork slot		298 mm / 8.2 mm	348 mm / 10.2 mm

Drawn by sheet_AB.py from the parts library (CadQuery parts P-04, P-04A, P-10, P-10A). Not to scale between the two sections' text; the geometry is to scale. Question for Marco Cross and Rick Fleeter: do we keep leak before break (A) or take the view (B)?